

Thirupathi Rao Murala

Java-First Full-Stack Developer | Spring Boot · Angular · Multi-Tenant SaaS · AI Integration

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PROFESSIONAL SUMMARY

Backend-first **Full-Stack Developer** with **5+ years** building and scaling features on a multi-tenant enterprise SaaS platform on **Java Spring Boot** and **Angular**. Own features end-to-end — from database design and REST APIs to production Angular UIs. First engineer to introduce **AI (OpenAI)** into the product — an AI clinical assistant and computer-vision features now used daily — and built a **multilingual voice-AI agent** (LangGraph, RAG). Delivered capabilities beyond the core product including a third-party **KYC identity-verification** integration and a **capital-expenditure approval platform (Spring Boot 3 / React)**. Led a product MVP solo and contributed to a large-scale **Angular v16→v19 migration**. Experienced with **multi-tenant SaaS, JWT security, JPA/Hibernate tuning and Redis caching**.

TECHNICAL SKILLS

Backend	Java, Spring Boot (2.x/3.x), Spring MVC, Spring Data JPA, Hibernate, JdbcTemplate, REST APIs, Microservices concepts, JWT Authentication, Multi-Tenant Architecture; Python / FastAPI (AI services)
Frontend	Angular 16–19, TypeScript, RxJS, HTML5, CSS3, Bootstrap 4/5, AdminLTE 3/4, Angular Material, PrimeNG, ng-bootstrap; React (Vite + TypeScript)
AI / Integration	OpenAI API (text & vision), Google Gemini, LangGraph agent orchestration, RAG, voice AI, prompt engineering, output validation & human-in-the-loop, third-party KYC integration, Google Maps API
Data & Caching	MySQL, SQL Server, PostgreSQL, SQL query optimization, indexing, schema design, Redis caching
DevOps & Tools	Git, AWS CodeCommit, Maven, Postman, Swagger / SpringDoc, JUnit, Mockito, Jasmine, Karma
Practices	End-to-end feature ownership, REST API design, SaaS multi-tenancy, code reviews, Agile delivery

WORK EXPERIENCE

Full-Stack Developer Aug 2020 – Present
Aititude IT Private Limited | Multi-Tenant Enterprise SaaS Hyderabad, India

- Introduced **AI to the product for the first time** — built an **AI clinical assistant** (OpenAI), **AI prescription templates**, and a **computer-vision feature** reading odometer & fuel-receipt data from field photos — engineered with **structured output, business-rule validation and human-in-the-loop review** so AI is verified before reaching clinical or financial records.
- Led a **product MVP end-to-end as sole developer**, delivering the full stack (database, Spring Boot APIs, Angular UI) from design to production.
- Delivered **20+ production features** across HR, fleet, health, invoice and security modules on a **multi-tenant SaaS platform**, building within its JWT-based tenant-isolation and RBAC (role/permission) framework.
- Built an **incident-management** module end-to-end (assignment groups, Kanban board, status/audit trail, reports) and a **field-patrol compliance report** with Google Maps GPS routing and PDF export — used for SLA-compliance audits.
- Tuned backend performance — resolved **N+1 query issues** with JOIN FETCH / projections and applied **Redis caching** for permissions, config and lookup data.
- Built fleet, attendance, payroll and invoice features — vehicle-unit mapping, ODO reports, daily attendance with calendar API, holiday/payroll policy CRUD and dynamic payments.
- Built a unified **KYC / identity-verification framework** integrating a third-party provider to verify **10+ Indian identity documents** — with name-matching, face verification and hardened provider-error handling.
- Worked as part of the team that **migrated the platform from Angular 16 to 19** (AdminLTE 4 + Bootstrap 5) — upgrading and regression-testing feature modules, using staged upgrades and production-build gating to avoid regressions.

Principal & Lecturer (Computer Science) 2011 – 2020
Suvidya Degree College Chityal, Telangana

- Led academic and administrative operations as Principal for 9 years — building the leadership, communication and ownership strengths I bring to engineering teams.
- Taught **Java, C and Visual Basic**, developing the deep programming fundamentals behind a deliberate, successful transition into full-time software development.

KEY PROJECTS

AI Clinical Assistant OpenAI Clinical AI assistant for doctors with contextual suggestions and AI prescription templates — the product's first AI feature, engineered with validation and doctor review before save.	AI Vision Data Capture OpenAI Computer-vision pipeline reading odometer and fuel-receipt data from field photos into validated, structured records — removing manual data entry.
Conversational Voice-AI Agent LLM · RAG Multilingual voice agent exploring agentic AI — LangGraph orchestration, real-time speech-to-speech and retrieval-augmented generation (RAG).	Capital-Expenditure Approval Platform Spring Boot 3 Multi-tenant capital-expenditure product — Spring Boot 3 / Java 17 (SQL Server) + React; configurable multi-level approval engine, budget tracking, and a tamper-evident, verifiable audit trail.

EDUCATION

M.Sc. in Computer Science — Chaitanya (Deemed to be University), Telangana 2018